Introduction

With ever improving 3D seismic coverage, data quality and interpretation techniques, the trend in static reservoir modelling is towards more direct constraining of 3D-model reservoir architecture by the seismic data. Whilst seismic constraining can result in more realistic, better quality reservoir architecture models, there are a number of key problems to overcome namely:

1) The limited vertical resolution of the seismic, i.e., not enough to resolve vertical facies and property trends seen in wells;
2) The limited ability to capture the shapes of sedimentary objects (e.g., channels and bars) by variograms;
3) The limited ability for geostatistics-based modelling algorithms to capture the spatial arrangements between facies in a geologically consistent manner (e.g., crevasse splays attached to channels etc)

This paper discusses a number of different approaches to reservoir architecture-model constraining by 3D seismic based on some actual field examples. It will be shown that the choice of an appropriate conceptual geological model is very important as it has a direct bearing on the best suitable and appropriate method to "hard-code" or "soft-code" the seismic constraints into the static model architecture and/or property distribution. It will also be shown how simple "close-the-loop" comparisons of average property maps backed out from the model against seismic attribute maps used to constrain the model in the first place, can be a good means to quality check the modelling results. The emphasis in this paper is on clastic reservoir settings with complex "labyrinth-type" reservoir architecture (Weber and Van Geuns, 1990) as the seismic constraining in such type of situations is less straightforward than in layer-cake settings with gradational property trends.

Overcoming vertical resolution issues

The limits of seismic resolution can cause severe and geologically unrealistic "blurring" of vertical facies trends away from wells when using a "hands-off" seismic-constrained co-simulation approach to facies modelling. To overcome this problem, a possible work-around is to combine vertical facies trends from the wells with lateral trend-maps derived from seismic attributes. If the facies assemblages and vertical trends differ across areas with seismic response (for example a channel-overbank situation where the channel domain contains sharp-based, fining up sands whereas the overbank exhibits thin sands in a coarsening up) a workflow where seismic attributes are used to define different "seismic facies domains", which are then trend-analysed and modelled separately, can provide good results. This is illustrated by a case study in a lower coastal plain setting (Figure 1) where seismic and wells indicate presence of a large, sand-prone valley cut into surrounding succession of shales and thin, sparse sandy creeks. Consistent with this conceptual geological model, valley-fill and overbank "facies domains" were" polygonised based on seismic attributes and then 3D modelled separately using facies proportions and trends seen in the wells. The rather abrupt "break" in modelled facies patterns between valley-cut and overbank domains is consistent with the implied erosional relation between those two domains. In settings where facies changes are thought to be gradational (e.g., a shoreface setting), abrupt breaks in the facies patterns that typically result from a "per-seismic-domain" modelling approach are undesirable and alternative techniques, for example external drift-kriging a Vshale property using an inversion-derived "distance from sand-line" property as secondary variable, can work well.

Generating realistic sedimentary body shapes

Variogram-based modelling algorithms are very suited to handle well-data rich settings especially where facies and/or property trends are gradational. However, such algorithms are less well suited to model the sediment body shapes typically seen in clastics (e.g., channel-levee arrangements, bars) which can lead to rather unrealistic outputs in case of a "hands-off" seismic-constrained co-simulation approach to facies modelling. To overcome this problem, an alternative is to revert to object modelling. However, some stringent and fairly "manual" constraining of the object modelling by interpreted amplitude domains and trendlines from seismic must still be done to give good results. Controlling the objects "seed probability" alone is not enough since not only the seed position but also the orientation of the objects must be made to honour the seismic. Again, the case study of Figure 1 is a good illustration of the application of this workflow. From the seismic attribute maps, trendlines were digitized and converted to a sedimentary trend azimuth map which was then used to control geobody directions in the model. Back-comparison of reservoir-average Net-to-Gross from the model and seismic attribute maps used to constrain the model suggest a good consistency between modelling results and seismic.

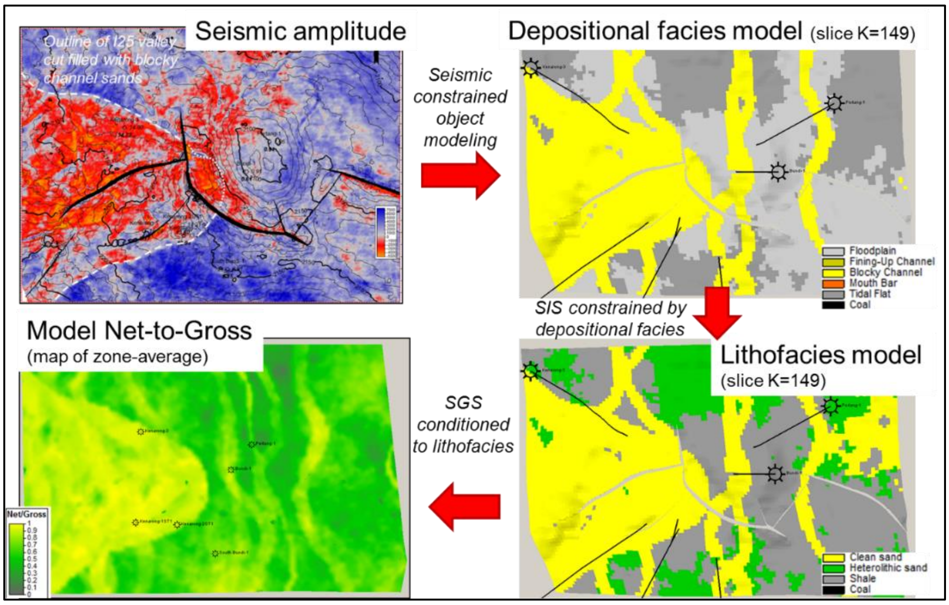

Figure 1: Case study of seismic-constrained reservoir architecture modelling in a lower coastal plain setting

Generating geologically consistent spatial facies arrangements

"Forcing" geostatistical algorithms to model, within the constraints of the data, spatial arrangements of different facies in a geologically consistent manner, e.g., crevasse splays attached to channels, levees getting sandier close to the feeder channel complex, can be difficult but is absolutely essential from a reservoir connectivity perspective. The use of nested workflows computing trend properties as intermediate outputs, for example the computing the distance to "primary" geological body surfaces or trendlines and then using this to control the population of "dependent" facies (e.g., splay facies spilling from a channel), is an elaborate but effective way of generating realistic facies patterns. Consider the case study of Figure 2, which is a deepwater channel-levee complex with scarce well control but good quality seismic, shows. First, individual channel elements were mapped from the seismic (physical picking the base-channel where possible, and then extending these to field-wide channel surfaces using trends mapped from spectral decomposition). The channel surfaces were then mapped into the 3D grid, honouring the time-stratigraphic relationships between the channels as observed from seismic. Then, a "normalized distance from channel-base" property was computed for each cell within the channel complex, whilst for the cells in the overbank domain a "distance from channel centre" property was computed. These "distance from" properties were in turn used to implement a fining-up facies trend in the channel complex, and a lateral grading of facies from proximal to distal levee in the overbank. Different conceptual models of the evolution of channel-levee complexes as derived from analogues, for example laterally grading or lateral and vertically grading levee complexes, and different assumptions on the sand versus mud-prone nature of later versus earlier channels, could also be built as geologically consistent reservoir model realizations.

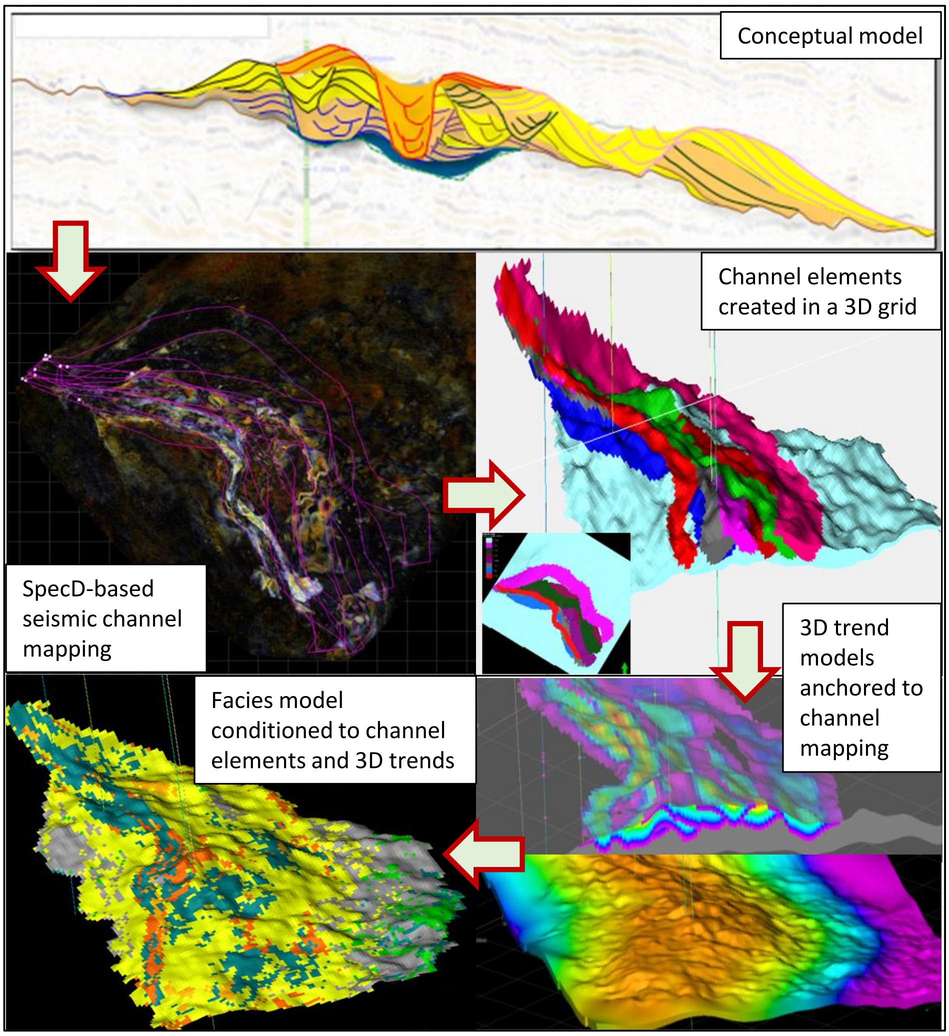

Figure 2: Case-study of seismic constrained reservoir architecture modelling for a deepwater channel-levee complex

Conclusions

The 3D seismic constraining of clastic reservoir-architecture models is best done in a more-or-less deterministic fashion by use of interpreted trendlines and/or seismic facies domains, rather than hands-off geostatistical modelling like co-simulation. Nested workflows involving deterministic mapping of objects from seismic and from there, backing out spatial and directional trends for constraining of facies and properties, can deliver good results and also allow for a geologically consistent set of model alternatives.

References

Weber, K.J. and Van Geuns, L. [1990]. Framework for constructing clastic reservoir simulation models. Journal of Petroleum Technology, **42**, p. 1248-1254